
Cloud Project

Image Rag Assistant

Image RAG ASSISTANT

Relax & Let AI Organize Your Memories





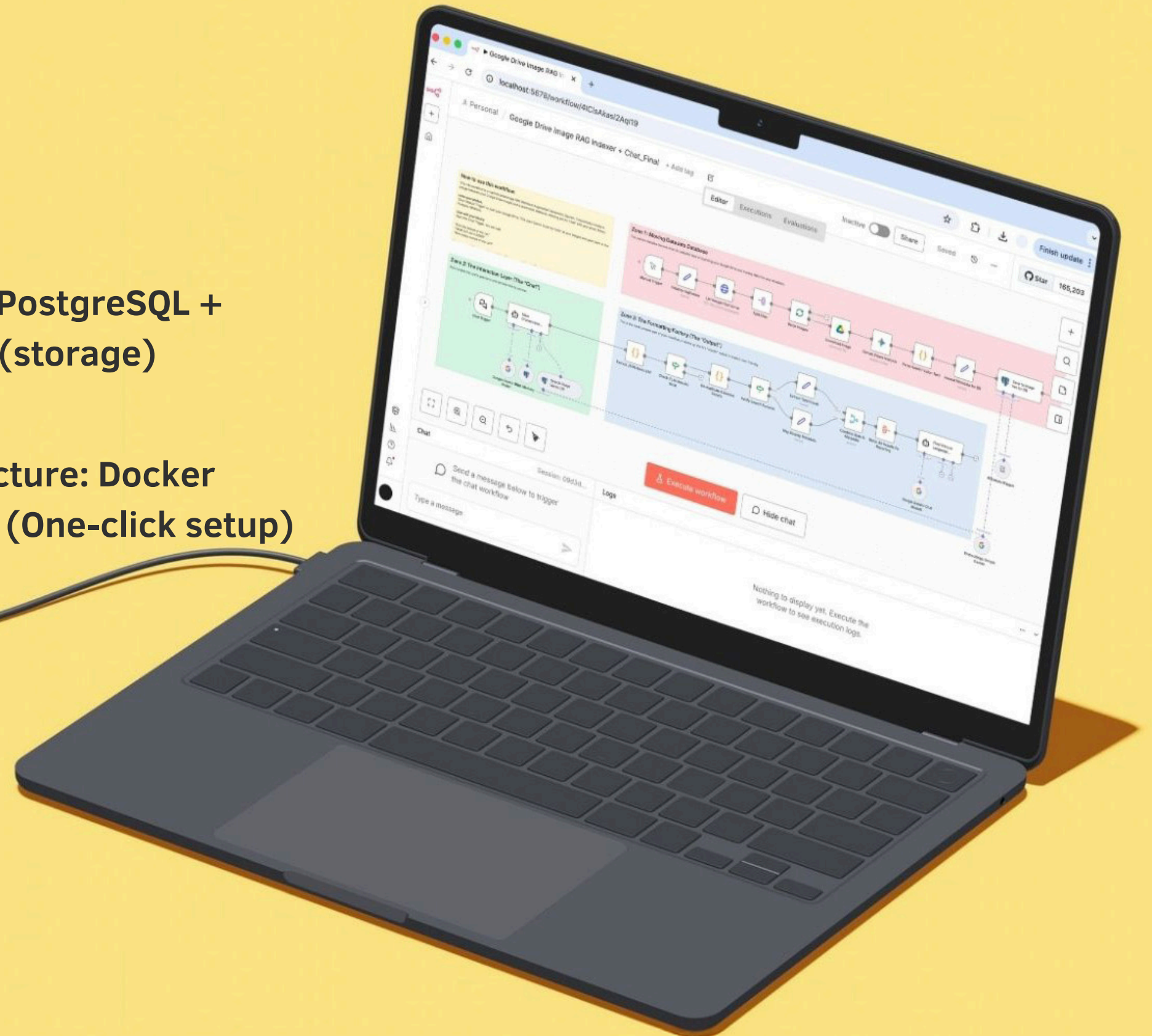
Core Problem

- Manual tagging is slow.
- Search is limited to dates and filenames.

Goal: Natural language retrieval and analysis.

Tech Stack

- ✓ **Orchestration: n8n**
(Low-code workflow engine)
- ✓ **Memory: PostgreSQL + pgvector** (storage)
- ✓ **Infrastructure: Docker Compose** (One-click setup)
- ✓ **Vision: Gemini 2.5 Flash**
(VLM & Embeddings).
- ✓ **Memory: PostgreSQL + pgvector** (Semantic storage).



System Overview

How to use this workflow:

This n8n workflow is a sophisticated Image RAG (Retrieval-Augmented Generation) System. It essentially creates a bridge between your Google Drive images and a searchable database, allowing you to "chat" with your photo library.

Index your photos:

Click "Manual Trigger" to scan your Google Drive. This uses Gemini Vision to "look" at your images and save them to the Postgres database.

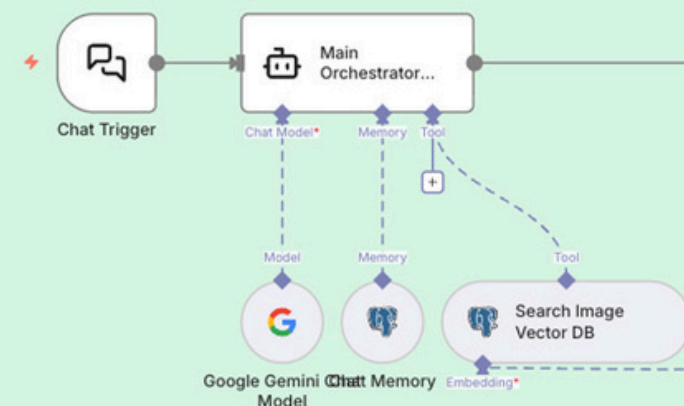
Chat with your library:

Open the Chat Trigger. You can ask:

"Find the photos of my cat."
"What did I do in 2020?"
"How many photos of my cat?"

Zone 2: The Interaction Layer (The "Chat")

This handles the user's questions and decides how to answer.



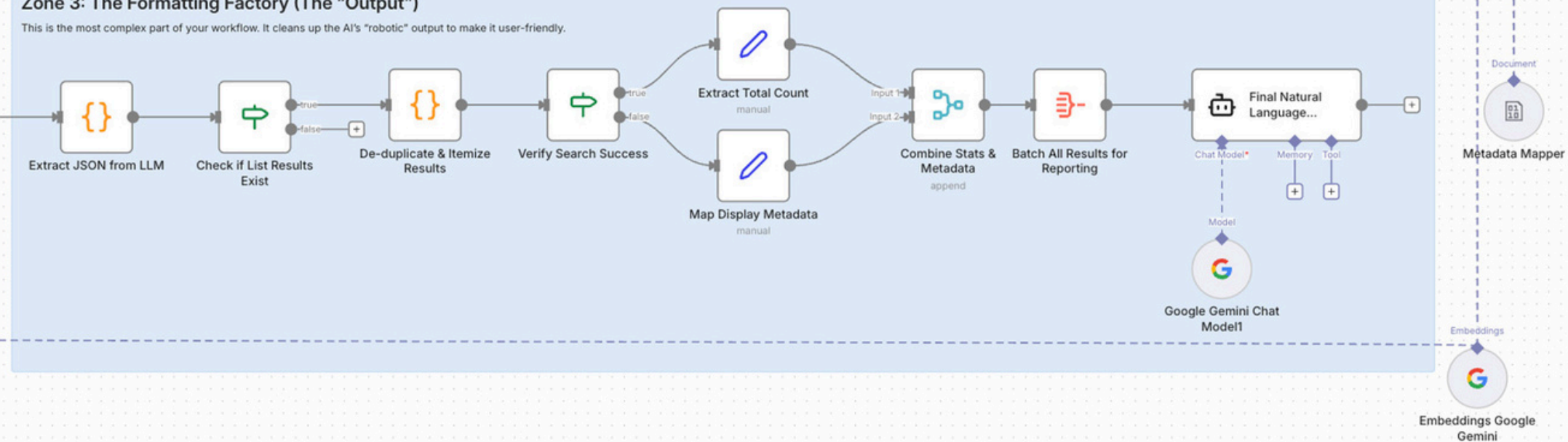
Zone 1: Moving Data into Database

This section handles the one-time (or periodic) task of scanning your Google Drive and moving data into your database.



Zone 3: The Formatting Factory (The "Output")

This is the most complex part of your workflow. It cleans up the AI's "robotic" output to make it user-friendly.



Pipeline

PIPELINE A: THE INDEXER

- Scan Google Drive.
- Gemini analyzes visual content.
- Convert descriptions into vectors.

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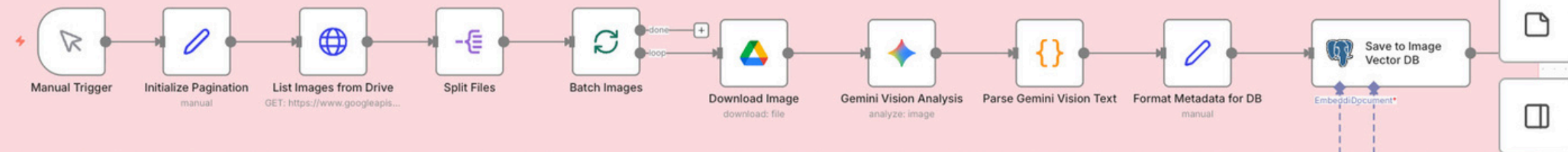
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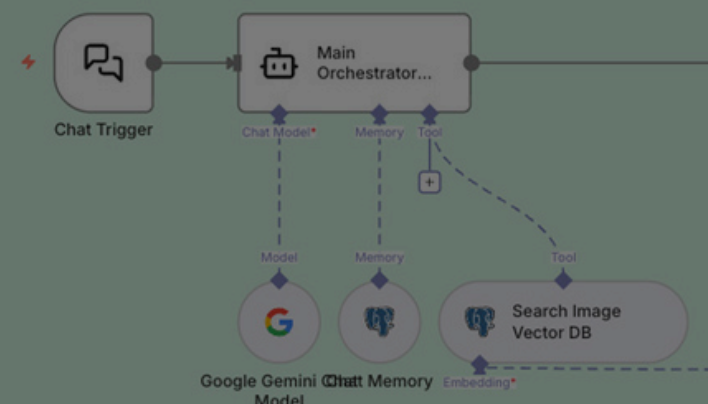
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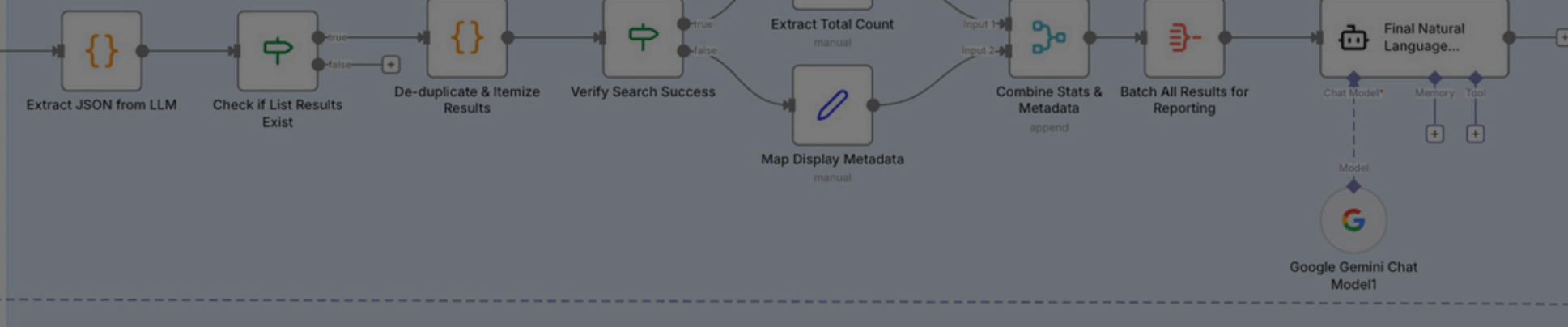
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Zone 3: The Formatting Factory (The "Output")

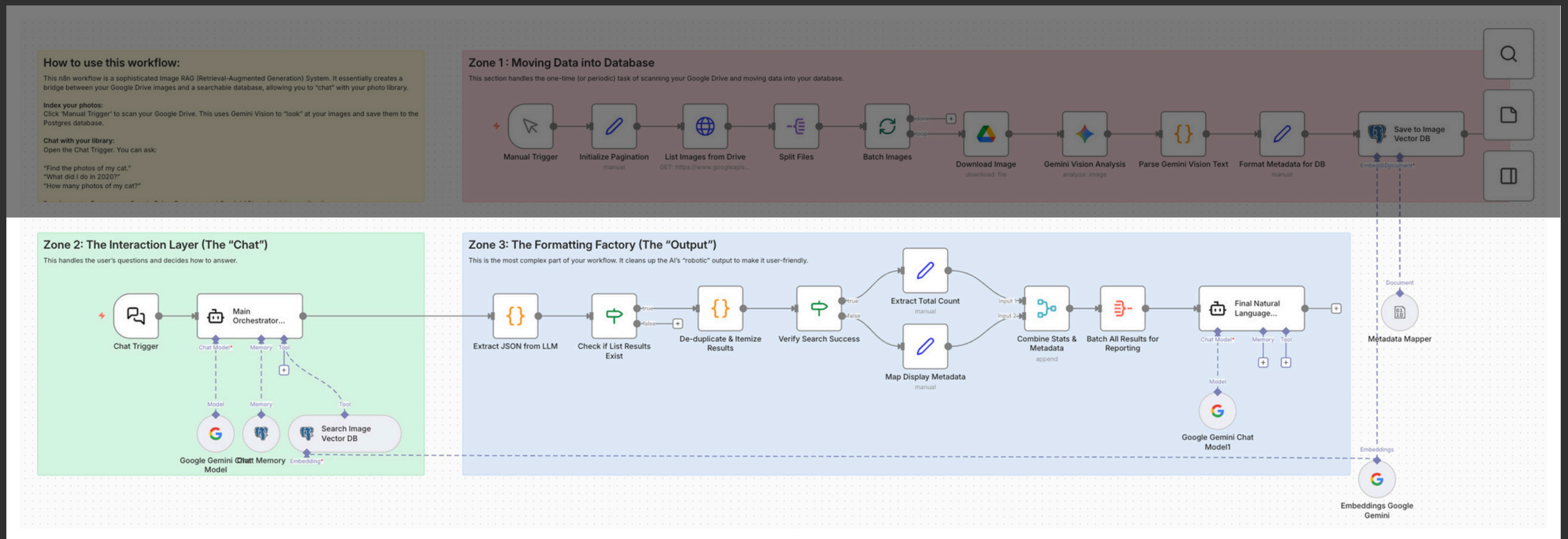
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Pipeline

PIPELINE B: THE ASSISTANT

- User asks a question via Chat.
- Agent performs a vector search.
- Gemini synthesizes a "Story" with direct links.



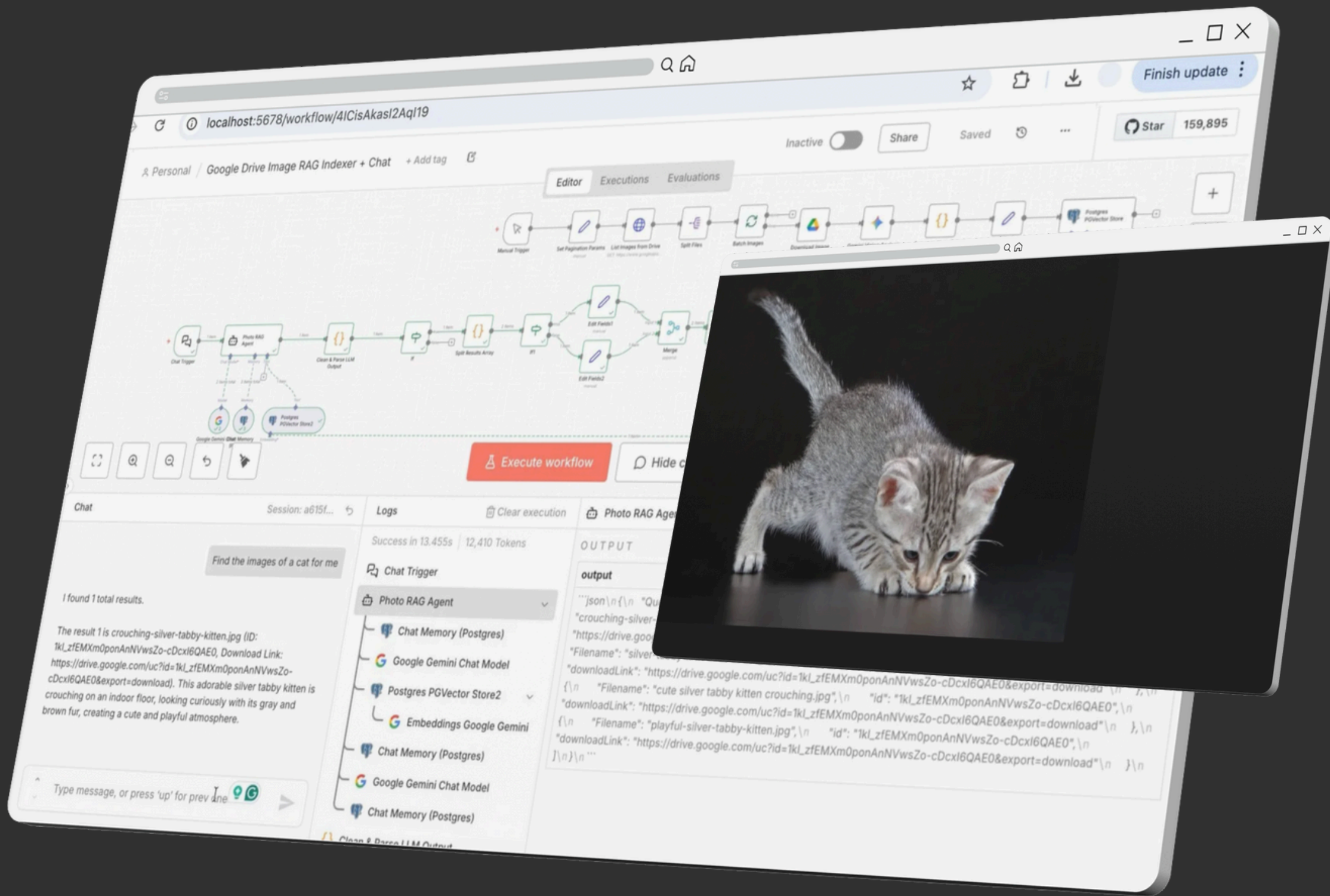
Project Structure

image-rag-assistant/

- ├── **docker-compose.yml** # Configuration to launch Google Drive, n8n, Gemini, and DB services.
- ├── .env.example
- ├── README.md
- ├── architecturediagram.pdf
- ├── .gitignore
- ├── n8n-workflows/
 - └── **gdrive-image-rag-final.json** # The core RAG workflow.



Demo Video



<https://youtu.be/DZQ9J00NTHQ>

09.

Implementation Guide

THREE STEPS TO LAUNCH

THE IMPLEMENTATION GUIDE

STEP 01: CONFIGURE

API keys in .env.

STEP 02: DEPLOY

docker-compose up.

STEP 03: IMPORT

Activate the n8n workflow.

10. Why This Matters

- **Automation:** Zero manual tagging required.
- **Intelligence:** Understands context
- **Scalability:** Works across any size Google Drive library.



Q&A

Thank You

Now let's start building something great.

11.

